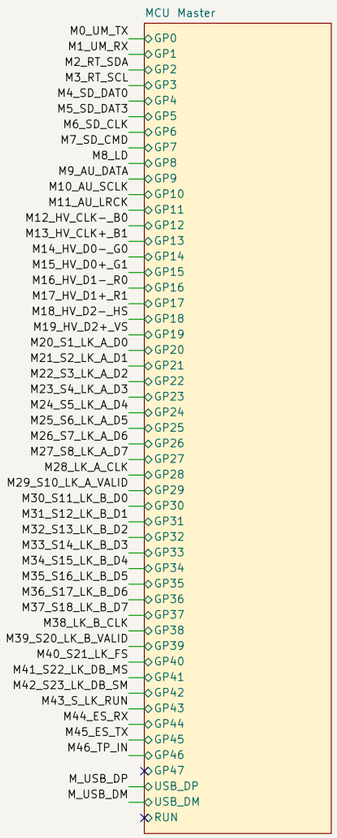
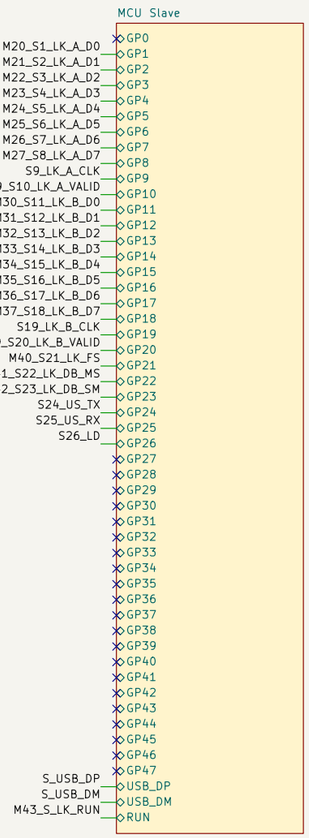
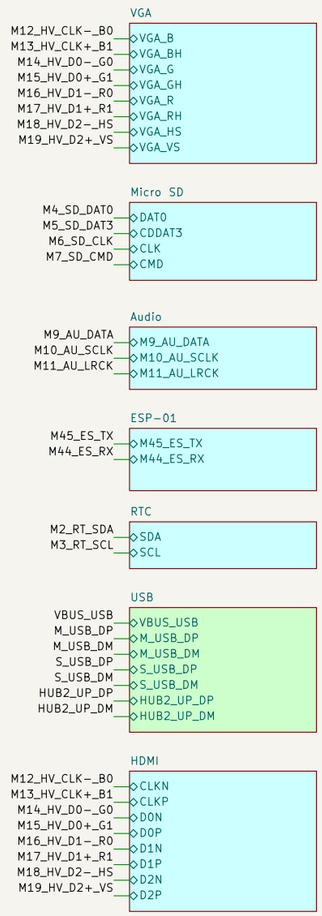
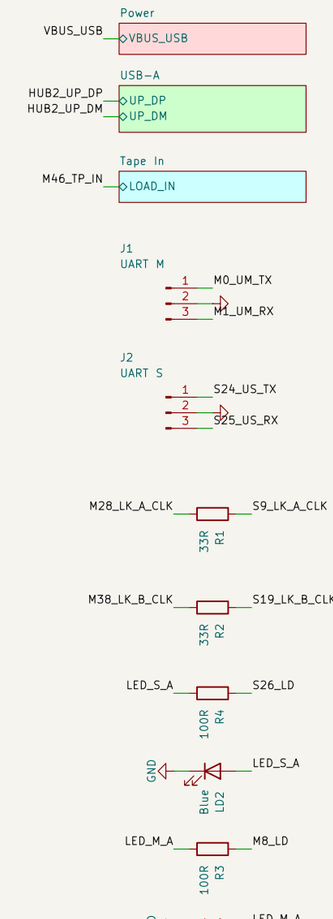
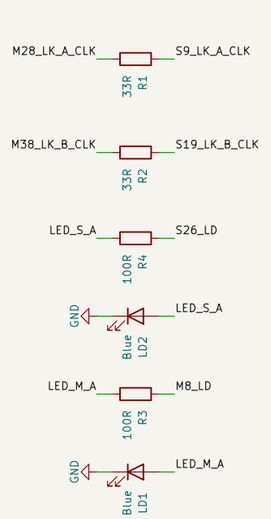










FRANK PGA2 — two PGA2350 (RP2350B) joined by the FRANK Core2 parallel link.

LINK (identical to frank_core2_proto, firmware compatible pin numbers):
Bus A D0..D7 master GP20..27 -> slave GP1..8. CLK master GP28 -> 33R -> slave GP9. VALID GP29 -> GP10
Bus B D0..D7 slave GP11..18 -> master GP30..37, CLK slave GP19 -> 33R -> master GP38, VALID GP20 -> GP39
FS GP40 -> GP21, DB_MS GP41 -> GP22, DB_SM GP23 -> GP42, master GP43 -> slave RUN
Place each 33R next to the DRIVING pin (R for bus A at the master, bus B at the slave).

MASTER: GP0/1 UART0 (J UART M), GP2/3 I2C1 -> DS3231 RTC, GP4-7 SD, GP8 STATUS LED (LD1), GP9-11 TDA1387, GP12-19 HDMI + VGA (shared, as FRANK PGA), GP44/45 UART0 <-> ESP-01, GP46 TAPE IN, USB -> USB sheet.
SLAVE: GP24/25 UART1 (J UART S), GP26 LED, USB -> hub port 1. GP27-46 free.
USB: hub 1 (P0 master in PC mode, P1 slave, P2 -> hub 2) ; hub 2 -> 4x USB-A.
GP47 = PSRAM CS inside PGA2350 (not used externally).
UART NOTE: master GP0/1 (J UART M) and GP44/45 (ESP-01) are BOTH UART0 on the RP2350 — only one can use the hardware UART at a time; run the other one as a PIO UART in firmware.

Differences vs Core2 proto firmware: slave PSRAM is the PGA2350 one (CS GP47, not GP0); master GP46 = TAPE IN; master status LED on GP8 (LD1); tape on FRANK PGA was GP22.

H1 Hole 2.5mm
H2 Hole 2.5mm
H3 Hole 2.5mm
H4 Hole 2.5mm

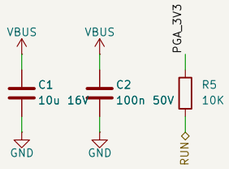
Nets: <M|S><GPIO>_<block>_<signal>. Blocks: SD HV AU TP ES RT UM US LK LD

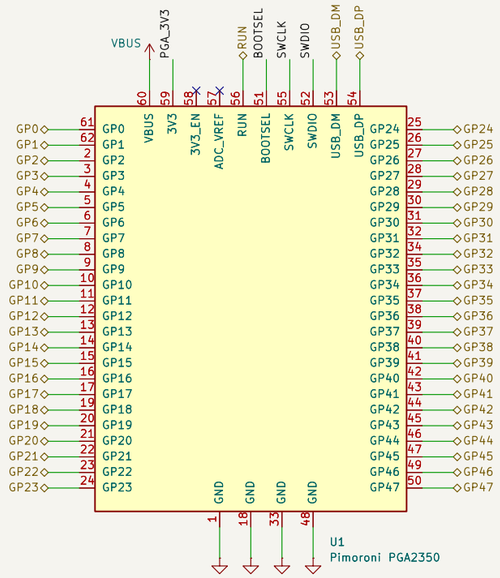
FRANK PGA2

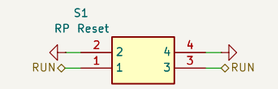
Sheet: /
File: frank_pga2.kicad_sch

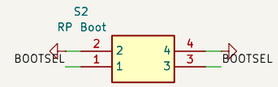
Title: FRANK PGA2 — dual PGA2350

Size: A3	Date: 2026-10-27	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 1/15











PGA2350: RP2350B + 16MB flash + 8MB PSRAM (PSRAM CS = GP47 on the module; leave GP47 free).
Module is fed from VBUS (5V); its own 3V3 regulator powers the RP2350.
RUN pull-up goes to the module's own 3V3 (pin 59) so RUN never back-feeds an unpowered module.
This sheet is used twice: MCU Master and MCU Slave (slave RUN is also driven by master GP43).
DEBUG: JST-SH 3 pin, same pinout as FRANK PGA / Core2 (1 SWDIO, 2 GND, 3 SWCLK).
Place C (10u + 100n) close to the VBUS pin of the module.

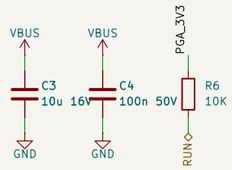
M = master GPIO, S = slave GPIO; UM/US = UART headers, LD = slave LED

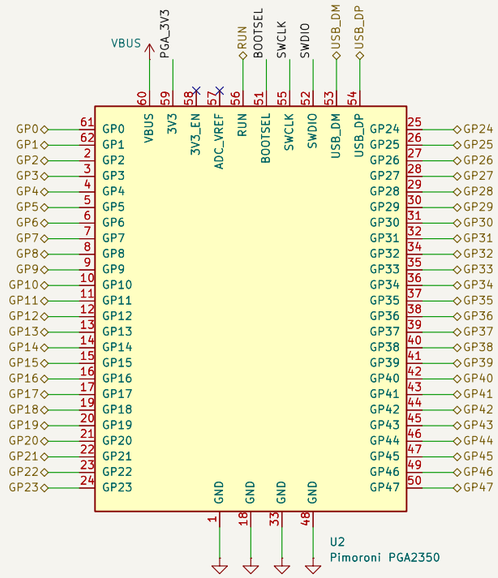
FRANK PGA2

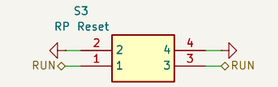
Sheet: /MCU Master/
File: pga2350.kicad_sch

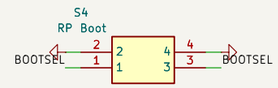
Title: PGA2350 module (Master / Slave)

Size: A3	Date: 2026-10-27	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 2/15











PGA2350: RP2350B + 16MB flash + 8MB PSRAM (PSRAM CS = GP47 on the module; leave GP47 free).
Module is fed from VBUS (5V); its own 3V3 regulator powers the RP2350.
RUN pull-up goes to the module's own 3V3 (pin 59) so RUN never back-feeds an unpowered module.
This sheet is used twice: MCU Master and MCU Slave (slave RUN is also driven by master GP43).
DEBUG: JST-SH 3 pin, same pinout as FRANK PGA / Core2 (1 SWDIO, 2 GND, 3 SWCLK).
Place C (10u + 100n) close to the VBUS pin of the module.

M = master GPIO, S = slave GPIO; UM/US = UART headers, LD = slave LED

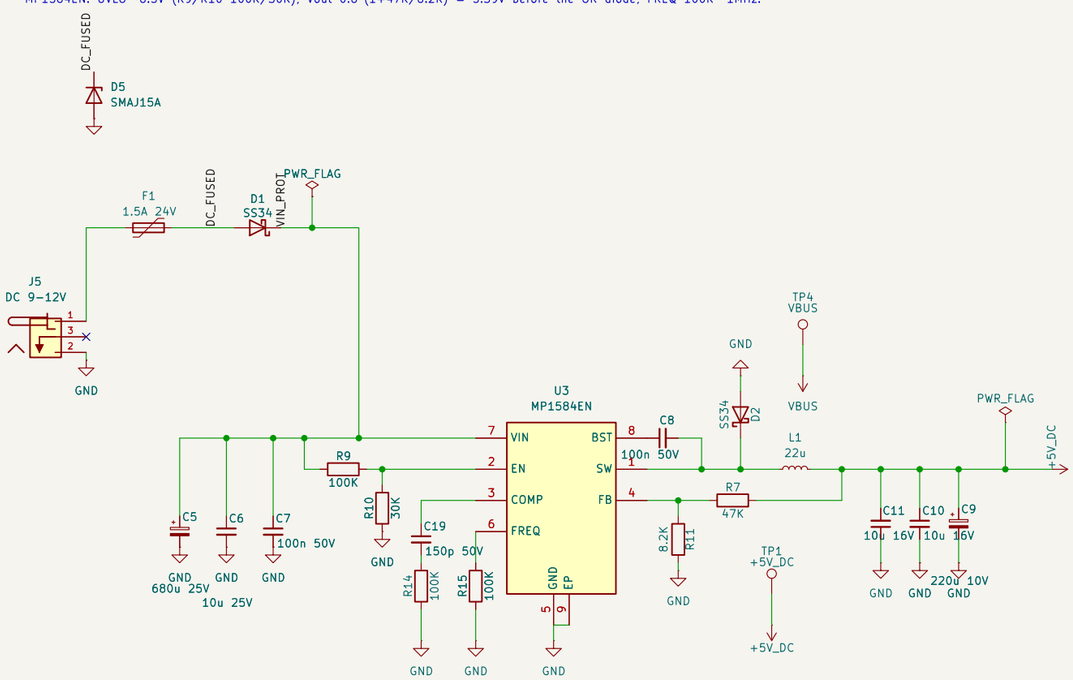
FRANK PGA2

Sheet: /MCU Slave/
File: pga2350.kicad_sch

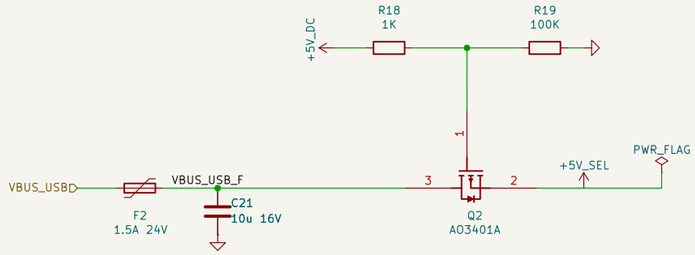
Title: PGA2350 module (Master / Slave)

Size: A3	Date: 2026-10-27	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 3/15

DC IN 9–12V (centre +, max 15V: SMAJ15A D5 starts conducting at ~16.7V). F1 PTC 1812, D1 SS34 reverse-polarity protection (a diode, not a P-FET, so the buck stops when the plug is pulled and the USB FET takes over once +5V_DC collapses).
MP1584EN: UVLO ~6.5V (R9/R10 100K/30K), Vout 0.8*(1+47K/8.2K) = 5.39V before the OR diode, FREQ 100K ~1MHz.

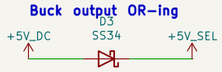


USB-C 5V input: auto-select P-FET
(off while the buck is running)
USB-C power: 1.5A max for the whole board (F2 hold)

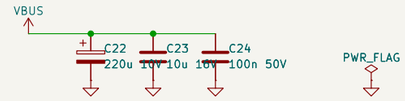


Q2 gate = +5V_DC (R18 1K / R19 100K): Q2 stays off while the buck runs and turns on only after +5V_DC has collapsed — no back-feed into the PC on DC unplug, and USB takes over if DC is present but below UVLO.

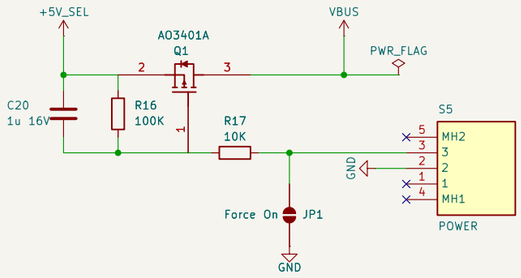
Buck output OR-ing

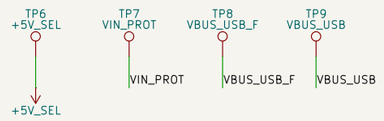


VBUS (switched 5V) bulk and decoupling

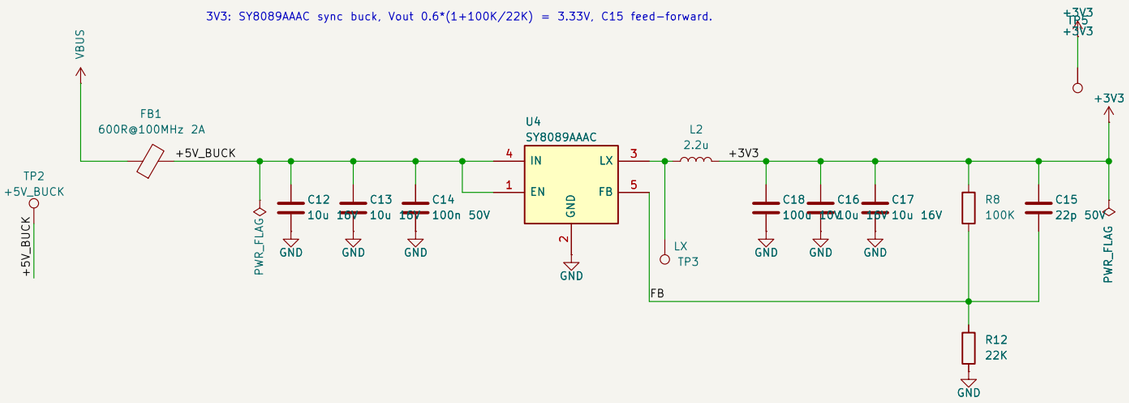


Power switch: Q1 high-side, S5 only pulls the gate; R16/C20 soft start; JP1 "Force On".

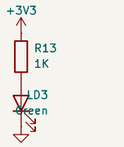




3V3: SY8089AAAC sync buck, Vout 0.6*(1+100K/22K) = 3.33V, C15 feed-forward.



3V3 power LED

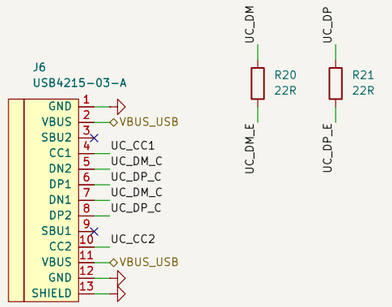


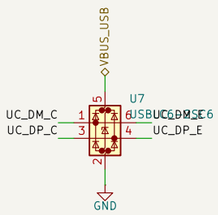
FRANK PGA2
Sheet: /Power/
File: power.kicad_sch

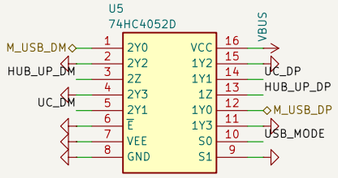
Title: Power: 9–12V DC / USB-C 5V, protection, 5V switch, 3V3 buck

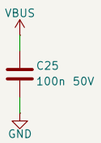
Size: A3	Date: 2026-10-27	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 4/15

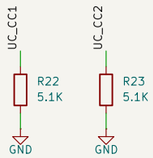
USB-C: power input (5V) + data. CC 5.1K = sink. 22R + USBLC6 ESD (same as FRANK PGA).
VBUS_USB goes to the Power sheet (PTC + auto source select).

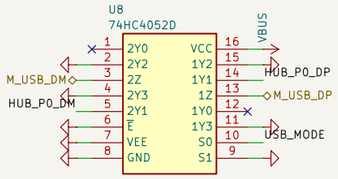


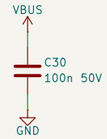


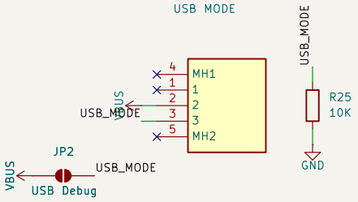






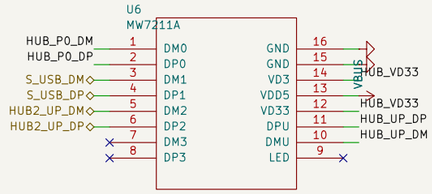


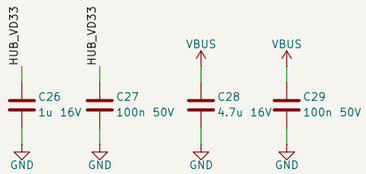




USB MODE switch (both 74HC4052D share S0, S1 = GND):
RUN (S0=0): master USB -> hub upstream. Master is USB host: keyboard/mouse on USB-A + slave on hub.
PC (S0=1): USB-C -> hub upstream. PC sees master (hub port 0, via mux B) and slave (hub port 1)
-> both RP2350 can be flashed (BOOTSEL) / used as CDC from one USB-C cable.
Mux B Y0 is left OPEN on purpose (master USB must not be loaded in RUN mode).

MW7211A hub 1: P0 = master (PC mode), P1 = slave RP2350, P2 -> hub 2 (USB-A ports), P3 spare.





FRANK PGA2

Sheet: /USB/
File: usb.kicad_sch

Title: USB: USB-C (power + data), 2x 74HC4052 mode mux, MW7211A

Size: A3	Date: 2026-10-27	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 5/15

Hub 2 (cascaded on hub 1 P2) drives all four USB-A ports (two stacked SS-52100-004).
Each port: CH217K current-limited switch (as dino_z80). Red LED = over-current on any port.
Power budget: 4 ports x 0.5A = 2A on top of ~0.8A for the rest (CH217K limit ~0.6A per port, R = 100K).
Full 4-port load: 12V / 2A adapter (9V needs ~1.7A, above the 1.5A hold of the DC input PTC F1).
On USB-C power the whole board is limited to 1.5A (F2 hold): light USB-A loads only (keyboard, mouse).

U9
MW7211A
P0_DM 1 DM0
P0_DP 2 DP0
P1_DM 3 DM1
P1_DP 4 DP1
P2_DM 5 DM2
P2_DP 6 DP2
P3_DM 7 DM3
P3_DP 8 DP3
GND 16
GND 15
VD3 14 → VBUS
VDD5 13 → HUB2_VD33
VD33 12 ← HUB2_VD33
DPU 11 ◇ UP_DP
DMU 10 ◇ UP_DM
LED 9 ✕

HUB2_VD33 → C31 1u 16V → GND
HUB2_VD33 → C32 100n 50V → GND
VBUS → C33 4.7u 16V → GND
VBUS → C34 100n 50V → GND

U19 VBUS_A1
UA1_DM 1 ~ 6 UA1_DM
UA1_DP 3 ~ 4 UA1_DP
5 VBUS_A1
2 GND
U20 VBUS_A2
UA2_DM 1 ~ 6 UA2_DM
UA2_DP 3 ~ 4 UA2_DP
5 VBUS_A2
2 GND
U21
UA4_DM 1 ~ 6 UA4_DM
UA3_DM 3 ~ 4 UA3_DM
5 ✕
2 GND
U22
UA3_DP 1 ~ 6 UA3_DP
UA4_DP 3 ~ 4 UA4_DP
5 ✕
2 GND

P0_DP → R26 22R → UA1_DP
P0_DM → R27 22R → UA1_DM
P1_DP → R28 22R → UA2_DP
P1_DM → R29 22R → UA2_DM
P2_DP → R30 22R → UA3_DP
P2_DM → R31 22R → UA3_DM
P3_DP → R32 22R → UA4_DP
P3_DM → R33 22R → UA4_DM

J7
SS-52100-004
VBUS1 1 VBUS_A1
VBUS2 5 VBUS_A2
D1+ 3 UA1_DP
D1- 2 UA1_DM
D2+ 7 UA2_DP
D2- 6 UA2_DM
Shield 9, GND1 4, GND2 8 → GND GND GND

U10
CH217K
VBUS_A1 1 VOUT
USB_OC 2 GND
3 FLAG#
VIN 6 VBUS
ISET 5 UA_ISET1
EN# 4
UA_ISET1 → R34 100K → GND
VBUS_A1 → C35 100u 10V → GND
VBUS_A1 → C36 100n 50V → GND

U11
CH217K
VBUS_A2 1 VOUT
USB_OC 2 GND
3 FLAG#
VIN 6 VBUS
ISET 5 UA_ISET2
EN# 4
UA_ISET2 → R37 100K → GND
VBUS_A2 → C37 100u 10V → GND
VBUS_A2 → C38 100n 50V → GND

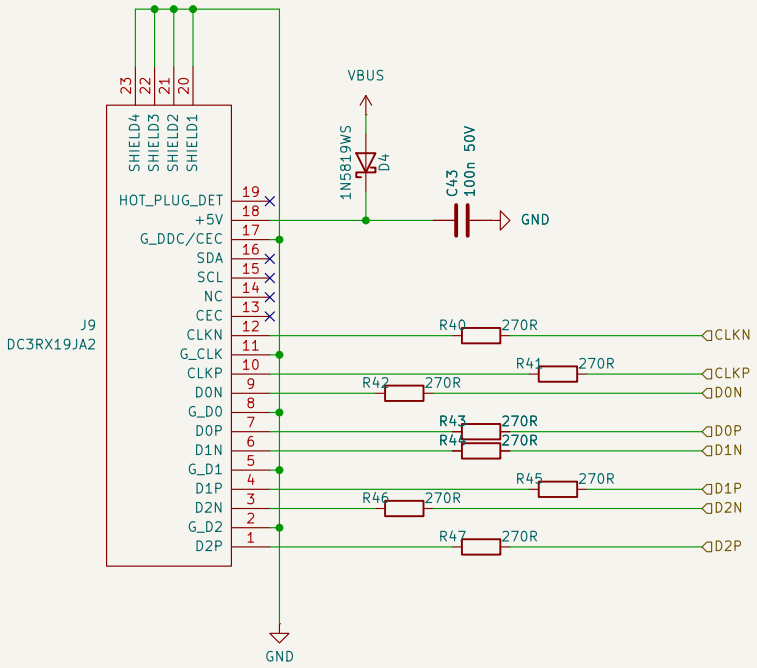
J8
SS-52100-004
VBUS1 1 VBUS_A3
VBUS2 5 VBUS_A4
D1+ 3 UA3_DP
D1- 2 UA3_DM
D2+ 7 UA4_DP
D2- 6 UA4_DM
Shield 9, GND1 4, GND2 8 → GND GND GND

U12
CH217K
VBUS_A3 1 VOUT
USB_OC 2 GND
3 FLAG#
VIN 6 VBUS
ISET 5 UA_ISET3
EN# 4
UA_ISET3 → R38 100K → GND
VBUS_A3 → C39 100u 10V → GND
VBUS_A3 → C40 100n 50V → GND

U13
CH217K
VBUS_A4 1 VOUT
USB_OC 2 GND
3 FLAG#
VIN 6 VBUS
ISET 5 UA_ISET4
EN# 4
UA_ISET4 → R39 100K → GND
VBUS_A4 → C41 100u 10V → GND
VBUS_A4 → C42 100n 50V → GND

VBUS → R35 100K → USB_OC
VBUS → R36 1K → OC_LED_A → LD4 Red → USB_OC

FRANK PGA2
Sheet: /USB-A/
File: usb_a.kicad_sch
Title: USB-A: MW7211A hub 2, 2x stacked USB-A, CH217K port power
Size: A3 | Date: 2026-10-27 | Rev: 1.00
KiCad E.D.A. 10.0.1 | Id: 6/15

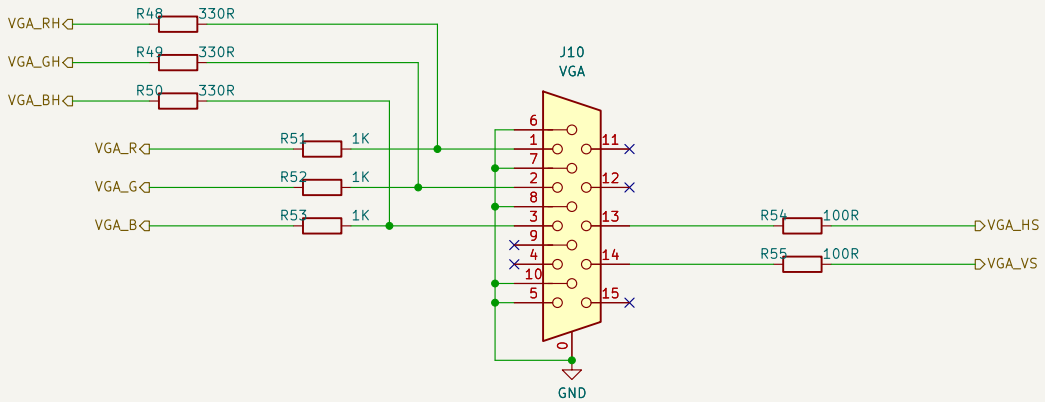


Block code: HV
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /HDMI/
File: hdmi.kicad_sch

Title: FRANK PGA2

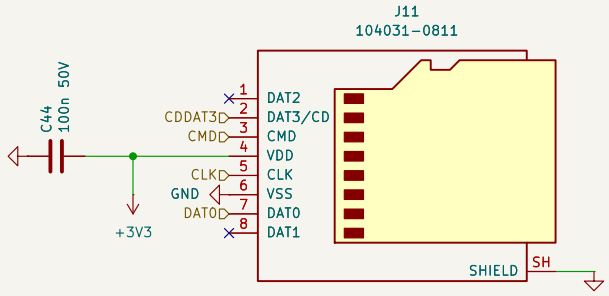
Size: A4	Date: 2026-10-29	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 7/15



Block code: HV
https://github.com/rh1tech/frank
Mikhail Matveev
Sheet: /VGA/
File: vga.kicad_sch

Title: FRANK PGA2

Size: A4	Date: 2026-10-29	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 8/15

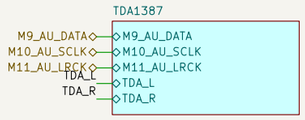


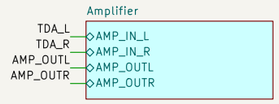
Block code: SD
https://github.com/rh1tech/frank
Mikhail Matveev

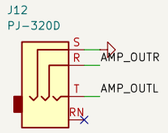
Sheet: /Micro SD/
File: sd.kicad_sch

Title: FRANK PGA2

Size: A4	Date: 2026-10-29	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 9/15







TDA1387 (same as FRANK PGA) -> LM358 line driver (same as FRANK Core2) -> 3.5mm jack.
No PWM audio on this board.

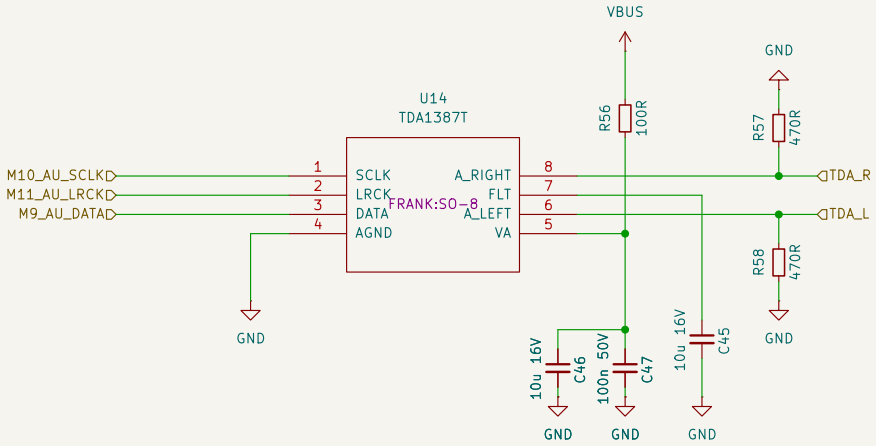
Block code: AU

FRANK PGA2

Sheet: /Audio/
File: audio.kicad_sch

Title: Audio: TDA1387 I2S DAC + LM358 line driver

Size: A3	Date: 2026-10-27	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 10/15

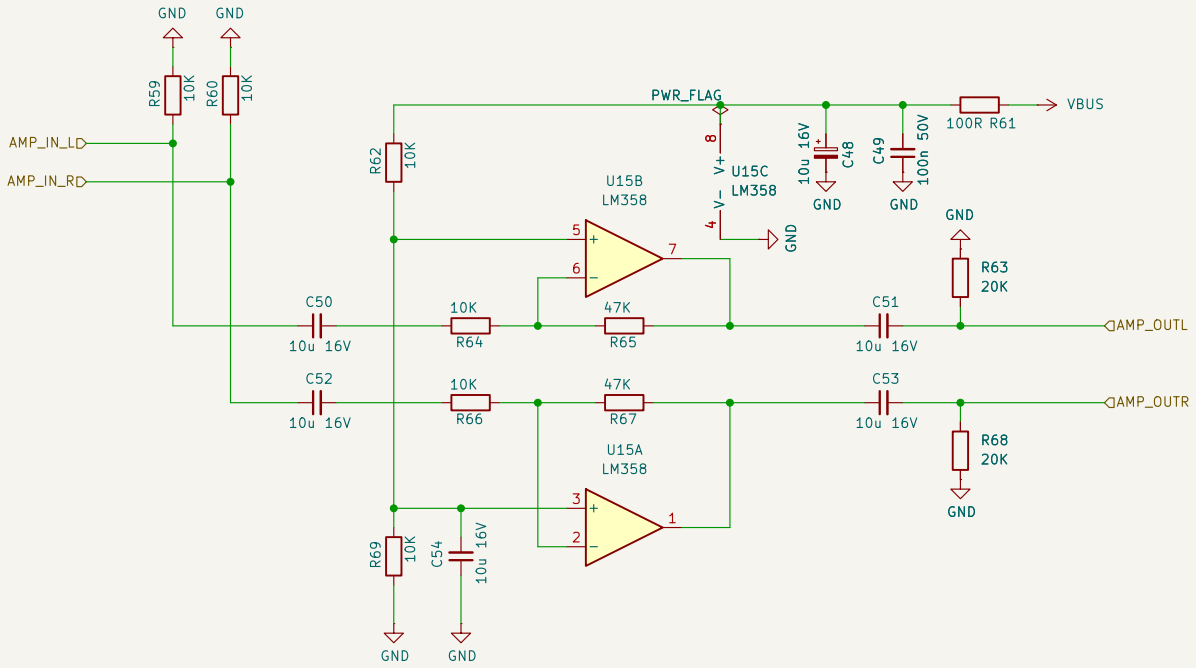


Block code: AU
https://github.com/rh1tech/frank
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Sheet: /Audio/TDA1387/
File: tda.kicad_sch

Title: FRANK PGA2

Size: A4	Date: 2026-10-29	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 11/15

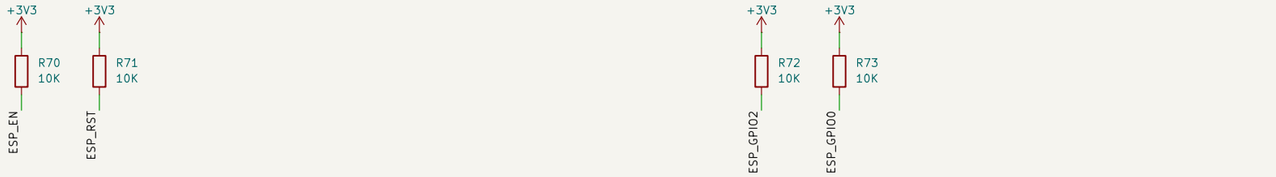


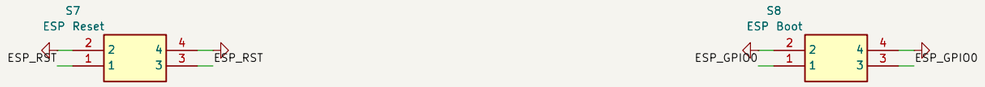
Block code: AU
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Audio/Amplifier/
File: amp.kicad_sch

Title: FRANK PGA2

Size: A4	Date: 2026-10-29	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 12/15





ESP-01 plugs into a 2x4 2.54mm female socket (footprint ESP-01S).
UART: master GP44 (UART0 TX) -> ESP RX, ESP TX -> master GP45 (UART0 RX).
GP0/1 (debug header J UART M) is also UART0: firmware runs one of the two as a PIO UART.
ESP RST is button-only (S7 + R71 pull-up); master GP8 drives the master status LED. ESP flashing: hold ESP Boot (GPIO0 low) and press ESP Reset. Pull-ups keep it in normal boot when the RP2350 pin floats.
100u + 10u + 100n near the socket for TX bursts.

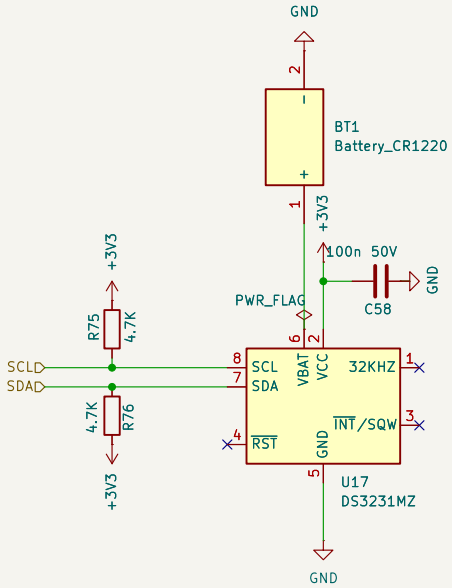
Block code: ES

FRANK PGA2

Sheet: /ESP-01/
File: esp01.kicad_sch

Title: ESP-01 / ESP-01S socket (WiFi, UART0 of master)

Size: A3	Date: 2026-10-27	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 13/15

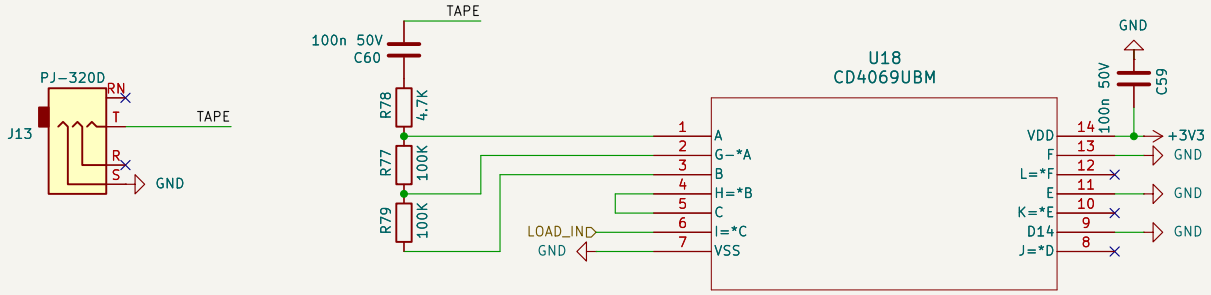


Block code: RT
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /RTC/
File: rtc.kicad_sch

Title: FRANK PGA2 PGA2

Size: A4	Date: 2026-10-29	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 14/15



Block code: TP
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Tape In/
File: tape.kicad_sch

Title: FRANK PGA2

Size: A4	Date: 2026-10-29	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 15/15